Type: GE3 / OE3 (Generic / Open Elective) B.Sc(ECS)-II (Semester III) Course Title: Operating Systems (Paper Code:)		
Credits: Theory – (2)		
Total Lectures: 30 Hrs.		**Contact Hrs. (L) :** 2
University Evaluation: 30 Marks		**Internal Evaluation:** 20 Marks
Course Outcomes:		
1.	To understand the main components of an OS and their functions.	
2.	To describe the functions of a modern OS concerning convenience, efficiency, and the ability to evolve.	
3.	To make aware of different types of OS and their services.	
4.	To learn different process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.	
Unit	**Content**	**Lect.**
I	Operating system: Definition of operating system, Types of Operating Systems-Batch, Multiprogramming, Time Sharing, Real-Time, Distributed, Parallel, OS Services, System components, and System Calls. **Process Management:** Concept of Process, Process states, Process Control Block, Context switching, Operations on Process.	**10**
II	**Process Synchronization and Deadlocks:** **Scheduling-** Concept of Process Scheduling, Types of Schedulers, Scheduling criteria, Scheduling algorithms Preemptive and Non-preemptive, FCFS, SJF, Round Robin, Priority Scheduling, Multilevel Queue Scheduling, Multilevel- feedback Queue Scheduling. **Process Synchronization:** The Producer Consumer Problem, Race Conditions, Critical Section Problem, Semaphores, and Classical Problems of Synchronization: Reader-Writer Problem, Dinning Philosopher Problem.	**20**
Reference Books:		
1.	Operating System Concepts By Siberchatz and Galvin.	
2.	Modern O.S. By Andrews Tanenbaum.	
3.	Operating System Concepts by Abraham Silberschatz, Peter B. Galvin, Greg Gagne.	